

Probability

Approaches and Parameter Estimation

Frequentist vs Bayesian

- Frequentists → prior assume করবে নেই না,
 - Treat parameters as fixed but unknown
 - Rely solely on observed data to make inferences
- Bayesian
 - Treat parameters as random variables with a defined prior
 - Updated beliefs with data to form a posterior distribution

এখন parameter আমাদের জ্ঞানভাণ্ডার আছে। একে knowledge আছে। যেখানে জায়গা থেকে না এসে, parameter space থেকে parameter গুলো আসবে। so এখন we have the prior knowledge. We have to update initial level value (after seeing the prior), so জ্ঞানভাণ্ডার.

এই parameter খুঁজা update করতে হবে মানে এ space থেকে আসবে।

Frequentist vs Bayesian



* এই random sample pick করা, But it can be anything. no guarantee that এই crowdlayf নাকি representative sample.

• Frequentists

- Training data is a random draw from the population
- Uncertainty in estimates is quantified through the sampling distribution
- Probability is **long-run frequency**

১০০ জন student $\Rightarrow$ ১০ জন pick
১০ জন কে কীভাবে pick করলাম তার উপর
আমাদের model টা highly dependent.
এভাবে sampling dataset এর উপর dependent
model হল এবং uncertainty চলে আসবে
আমরা coin-toss এর ক্ষেত্রে infinite
সময়ক বাবু করলে আমরা বুঝতে পারি, coin টা
Bias কিনা।

• Bayesian

- Uncertainty in estimates is quantified through the posterior distribution
- Probability is **degree of belief**

আমরা sampling distribution এর সাথে
uncertainty ধাক্কা চাইনা, তখন আমরা
আমরা যেখানে প্রত্যাশিত result কী হবে,
so assume করি, coin Bias. এখন ১০০ বার coin
toss করলাম। so coin toss করলে আমরা দেখলাম—

গাফা H, ৭৭৭৭ T পড়ল। so it is not actually biased
coin. Initial assumption কে এখন জাম্বা update কক্কা।

Bayes' Rule

- Given two events A and B,

$$P(A, B) = P(A|B)P(B) = P(B|A)P(A)$$

and,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Bayesian Approach

probability $\rightarrow$ degree of beliefs

- Given (training) data D and a parameter θ (*random variable*),
- Likelihood, $P(D|\theta)$
- Prior, $P(\theta)$
- Evidence, $P(D)$
- Posterior,

অন্য infinitely many θ পাচ্ছি

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Posterior $\propto$ Likelihood $\times$ Prior

Bayesian Approach

- Predictive distribution $\rightarrow$ expected value over all this θ .

$$P(y^*|x^*, D) = \int P(y^*|x^*, \theta)P(\theta|D)d\theta$$

- Integral over all values of θ using the posterior probability estimated earlier.

Distribution's support $\Rightarrow$ θ এর যেসব value এর জন্য distribution non-zero সেসবই হল distribution's support.

Coin Flipping

- Estimate the bias θ of a coin, i.e., the probability of heads, where $\theta \in [0, 1]$
- N flips of a coin has yielded k heads.

Coin Flipping

- Frequentist Approach
 - Coin flips are independent and identically distributed Bernoulli trials with unknown parameter θ
 - θ is fixed but unknown
- The likelihood of observing k heads out of n flips is:

$$L(\theta) = P(\text{data}|\theta) = \underbrace{\binom{n}{k} \theta^k (1 - \theta)^{n-k}}$$

$$L(\theta) = P(\text{data}|\theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

Coin Flipping

- Frequentist Approach

- We pick θ to maximize the likelihood
- If $\hat{\theta}_{MLE}$ is the value of θ that maximizes the likelihood then,

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

- Since $\binom{n}{k}$ doesn't depend on θ we can maximize the log likelihood as follows,

$$\log L(\theta) = k \log \theta + (n - k) \log(1 - \theta)$$

Coin Flipping

- Frequentist Approach

- Setting derivative w.r.t. θ to zero, we obtain,

$$\hat{\theta}_{MLE} = \frac{k}{n}$$

- MLE simply yields the empirical proportion of heads observed.
 - No priors about the bias of the coin is incorporated
 - Often confidence intervals are reported or hypothesis tests are performed

Coin Flipping

- Bayesian Approach

- The coin flips are i.i.d. Bernoulli trials with unknown bias θ . $\rightarrow$ θ is unknown
- Before observing data, we model θ as a **random variable** with a prior distribution reflecting our belief about θ .

- What will be our prior distribution? $\rightarrow$ θ is unknown prior distribution θ is unknown $\rightarrow$ θ is unknown $\rightarrow$ θ is unknown
- For ease of analysis, we shall pick a conjugate prior $\rightarrow$ posterior distribution θ is unknown $\rightarrow$ θ is unknown $\rightarrow$ θ is unknown

Conjugate Prior

- A prior distribution
 - When combined with a specific **likelihood function**, yields a **posterior distribution of the same family** as the prior.

Likelihood binomial distribution → যদি know হয় আরও → যদি জানবে লেটজনা চাই।

Likelihood (Data Model)	Parameter	Conjugate Prior	Posterior Distribution
Bernoulli / Binomial	θ	Beta(α, β)	Beta($\alpha + k, \beta + n - k$)
Poisson	λ	Gamma(α, β)	Gamma($\alpha + \text{sum}, \beta + n$)
Normal (known variance)	μ	Normal(μ, σ^2)	Normal (updated mean and variance)
Exponential	λ	Gamma	Gamma (updated parameters)

event
অনুঘট
Likelihood
স্থানো fixed
হবে।

$$L(\theta) = P(\text{data}|\theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

Coin Flipping

- Bayesian Approach

- The coin flips are i.i.d. Bernoulli trials with unknown bias θ .
- Before observing data, we model θ as a **random variable** with a prior distribution reflecting our belief about θ .

- Prior

- A conjugate prior for Bernoulli is the **Beta distribution**

$$\theta \sim \text{Beta}(\alpha, \beta)$$

- Likelihood

- Same as frequentist

Coin Flipping

- Bayesian Approach

- According to Bayes' theorem,

$$P(\theta|\text{data}) \propto P(\text{data}|\theta)P(\theta)$$

(posterior distribution) $\propto$ (known likelihood $\times$ prior)

- Since Beta prior distribution was used,

$$\theta|\text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

আমাদের α ও β কে
estimate করতে হবে।

Coin Flipping

- Bayesian Approach

- We have obtained a posterior distribution of θ

$$\theta|\text{data} \sim \text{Beta}(\alpha + k, \beta + n - k)$$

- To get the best estimate of θ following the posterior, we use Maximum A Posteriori (MAP) method

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} P(\theta|\text{data})$$

(MAP)

Coin Flipping

- Bayesian Approach
 - Make predictions about the next coin flip x_{new}

$$P(x_{new}|\text{data}) = \int P(x_{new}|\theta)P(\theta|\text{data})d\theta$$

Posterior Estimation → estimate/compute করা অসম্ভবমত intractable হয় মাত্র। Because,

- Posterior is intractable for most problems

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

→ এটা

compute করা intractable.

- How to do that?

- Approximate Inference (Later!!)

Week-7 & approximate inference
সাজানোরবে,